

Challenge (Habanero)

- Q1.** A hiker travels $\frac{3}{4}$ of a mile in $\frac{1}{2}$ hour. What is her speed in miles per hour?
(A) 1.5
(B) 2
(C) 2.5
(D) 3
(E) 3.5
- Q2.** A park ranger measures the population of deer in a forest. If the population increases by 25% from 120 deer to x deer, what is x ?
(A) 120
(B) 130
(C) 150
(D) 160
(E) 180
- Q3.** A car's fuel tank can hold $\frac{5}{8}$ of a gallon. If it currently has $\frac{3}{8}$ of a gallon, what fraction of the tank is empty?
(A) $\frac{1}{8}$
(B) $\frac{1}{4}$
(C) $\frac{1}{2}$
(D) $\frac{3}{4}$
(E) $\frac{5}{8}$
- Q4.** A river is $\frac{2}{3}$ mile wide. If a bridge spans $\frac{1}{4}$ of the river, what is the length of the bridge in miles?
(A) $\frac{1}{6}$
(B) $\frac{1}{4}$
(C) $\frac{1}{3}$
(D) $\frac{1}{2}$
(E) $\frac{2}{3}$
- Q5.** A bicycle travels $\frac{5}{6}$ of a mile on $\frac{1}{3}$ gallon of gas. How many miles will it travel on 1 gallon of gas?
(A) 1.5
(B) 2.5
(C) 3.5
(D) 4.5
(E) 5.5
- Q6.** A map is drawn to scale with 1 inch representing $\frac{1}{4}$ mile. If a distance on the map is 2.5 inches, what is the actual distance in miles?
(A) 0.5
(B) 0.625
(C) 0.75
(D) 1
(E) 1.25
- Q7.** A water tank can hold $\frac{3}{4}$ of a gallon. If $\frac{1}{8}$ of a gallon is already filled, what fraction of the tank is still empty?
(A) $\frac{1}{8}$
(B) $\frac{1}{4}$
(C) $\frac{1}{2}$
(D) $\frac{3}{4}$
(E) $\frac{5}{8}$
- Q8.** A group of friends want to share a trail mix that is $\frac{3}{5}$ nuts and $\frac{2}{5}$ dried fruit. If they have 20 ounces of trail mix, how many ounces of nuts do they have?
(A) 8
(B) 10
(C) 12
(D) 14
(E) 16

Open Ended Questions (FRQ)

- Q9.** A hiker is planning a trip that is 12 miles long. She wants to complete the trip in 4 hours. If she has already completed $\frac{3}{4}$ of the trip, what is her current speed in miles per hour? Show your work and explain your reasoning.
- Q10.** A park ranger is monitoring the population of a certain species of animal. The population is currently 120 and is increasing at a rate of 15% per year. If the park ranger wants to maintain a population of no more than 180, how many years can the population increase at this rate? Show your work and explain your reasoning.

Chef's Tips

- Read each question carefully and make sure to answer the question asked.
- Use a number line to help with fraction and decimal comparisons.
- Work carefully and check your calculations.